AN INVARIANT AFFINE OPEN AROUND GENERIC ORBIT POINTS CUT OUT BY AN ELEMENT OF I, FOR FINITE FINITELY PRESENTED GROUPOIDS WITH AN IMMERSION INTO AN AFFINE SCHEME

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Allowed standard prerequisites (editorial).

- Invariant affine opens around generic orbit points for groupoids with finite source and target (more-groupoids chapter, Lemma lemma-invariant-affine-open-around-generic-point, canonical tag 0ABC). cited at proof lines 2936-2937. Provides the neighbourhood V' of the orbit generic points inside V inside U on which the groupoid restricts to a finite locally free affine situation, enabling the replacement reduction.
- Prime avoidance style element choice avoiding finitely many points (algebra chapter, Lemma algebra-lemma-silly, canonical tag 00DS). cited at proof lines 2947-2949. Justifies picking f in I that does not vanish at the finitely many orbit points.
- The determinant trick making the norm of a pullback function R-invariant (groupoids chapter, Lemma groupoids-lemma-determinant-trick, canonical tag 03BH). cited at proof lines 2956-2958. Gives the R-invariance of g, hence of D(g), the core of the invariance conclusion.

Source reference. Original lemma 0ABC: sources/stacks-chow/tags.txt, source lines 13015–13015. Cited at proof lines 2936-2937: supplies V' around the generic orbit points and the finite locally free affine reduction.

Source reference. Original lemma 00DS: sources/stacks-chow/tags.txt, source lines 508–508. Cited at proof lines 2947-2949: justifies the choice of f in I not vanishing at the finitely many orbit points.

Source reference. Original lemma 03BH: sources/stacks-chow/tags.txt, source lines 4348–4348. Cited at proof lines 2956-2958: g is an R-invariant function, so D(g) is an R-invariant open.

Lemma 0.2. *Let (U, R, s, t, c) be a groupoid scheme with s, t finite and of finite presentation. Let $u_1, \ldots, u_m \in U$ be points whose R-orbits consist of generic points of irreducible components of U. Let $j : U \to \mathrm{Spec}(A)$ be an immersion. Let $I \subset A$*

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be an ideal such that $j(U) \cap V(I) = \emptyset$ and $V(I) \cup j(U)$ is closed in $\mathrm{Spec}(A)$*. Then there exists an $h \in I$ such that $j^{-1}D(h)$ is an R-invariant affine open subscheme of U containing $u_1, \ldots, u_m$.*

Proof. Let $u_1, \ldots, u_m \in V' \subset V \subset U$ be as in Lemma 0ABC. Since $U \setminus V$ is closed in U, j an immersion, and $V(I) \cup j(U)$ is closed in $\mathrm{Spec}(A)$, we can find an ideal $J \subset I$ such that $V(J) = V(I) \cup j(U \setminus V)$. For example we can take the ideal of elements of I which vanish on $j(U \setminus V)$. Thus we can replace (U, R, s, t, c), $j : U \to \mathrm{Spec}(A)$, and I by (V', R', s', t', c'), $j|_{V'} : V' \to \mathrm{Spec}(A)$, and J. In other words, we may assume that U is affine and that s and t are finite locally free. Take any $f \in I$ which does not vanish at all the points in the R-orbits of $u_1, \ldots, u_m$ (Algebra, Lemma 00DS). Consider

$$g = \mathrm{Norm}_s(t^\sharp(j^\sharp(f))) \in \Gamma(U, \mathcal{O}_U)$$

Since $f \in I$ and since $V(I) \cup j(U)$ is closed we see that $U \cap D(f) \to D(f)$ is a closed immersion. Hence $f^n g$ is the image of an element $h \in I$ for some $n > 0$. We claim that h works. Namely, we have seen in Groupoids, Lemma 03BH that g is an R-invariant function, hence $D(g) \subset U$ is R-invariant. Since f does not vanish on the orbit of u_j, the function g does not vanish at u_j. Moreover, we have $V(g) \supset V(j^\sharp(f))$ and hence $j^{-1}D(h) = D(g)$. □